Does seasonality help forecast fashion search interest?

Indian and French markets | LSTM vs seasonal naive | Google Trends 2016–2026

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Fashion attention is seasonal and local

Festivals, weather and retail calendars shape what people look for in India and in France.

Search data is a timely, cheap signal

Google Trends indices are used to nowcast demand-related activity (Choi & Varian, 2012).

LSTMs are popular, but is extra seasonal input needed?

Neural nets are not automatically better than simple benchmarks (Hewamalage et al., 2021).

Data are independently normalized search-interest indices, not sales. Refs: Choi & Varian, Econ. Record 88(s1), 2012; Hewamalage et al., Int. J. Forecasting 37(1), 2021.

Research gap and objectives

Gap

Few controlled comparisons of seasonality-aware vs standard LSTMs on search-interest series from an Indian and a European market (within the sources we reviewed).

Our contribution: a controlled comparison of cyclical calendar features, a seasonal naive baseline and anomaly sensitivity, not a new model.

1. Do LSTMs beat a seasonal naive benchmark?
2. Do sine/cosine month features improve a standard LSTM?
3. Do conclusions differ between India and France?

Dataset and methodology

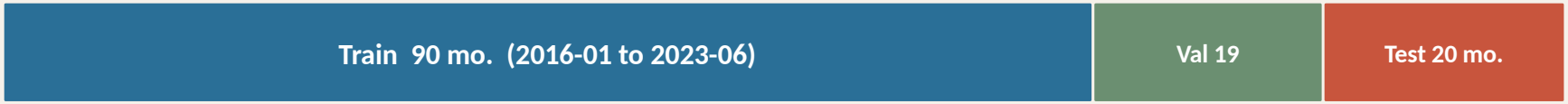
Google Trends, monthly, Jan 2016 – Sep 2026

- 4 keywords: streetwear, oversized clothing, sustainable fashion, traditional clothing
- 2 countries: India, France = 8 series
- 129 months each; 0 missing, 0 duplicates, gap-free

Data caveats

- Index (peak = 100 per series), not volume or sales; series normalized independently
- All 8 series hit 100 in Feb 2026 and shift up in Nov 2025 (unexplained); reported with and without
- France oversized: 98% zeros

Chronological split per series (identical test months for all models)



Leakage control: scaler fitted on training data only; backward-looking 12-month windows; validation used only for early stopping; one-step walk-forward test (Feb 2025 – Sep 2026).

Models and experimental setup

Seasonal naive

Forecast = value 12 months earlier.

$y(t) = y(t-12)$

Standard LSTM

12 lagged values in; LSTM(50) → Dense(1).

input: 12 × 1 feature

Seasonality-aware LSTM

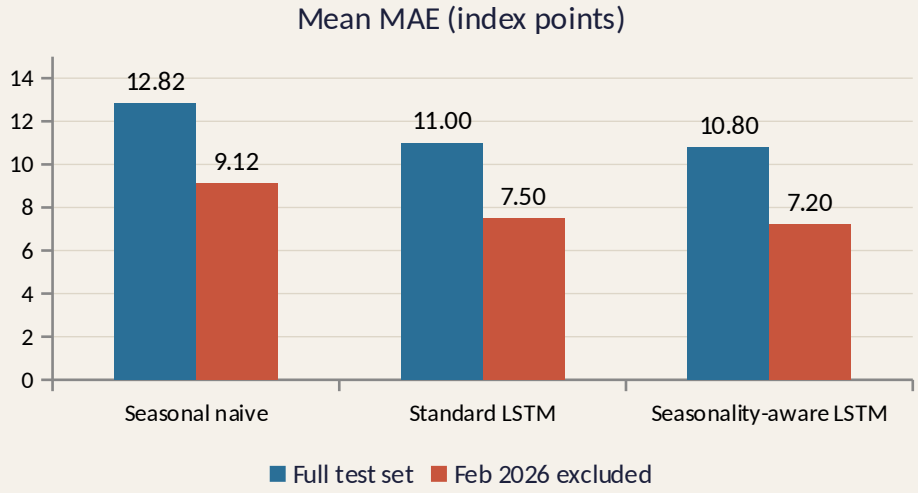
Same network plus month encoding of the target month.

input: 12 × 3 (value, sin, cos)

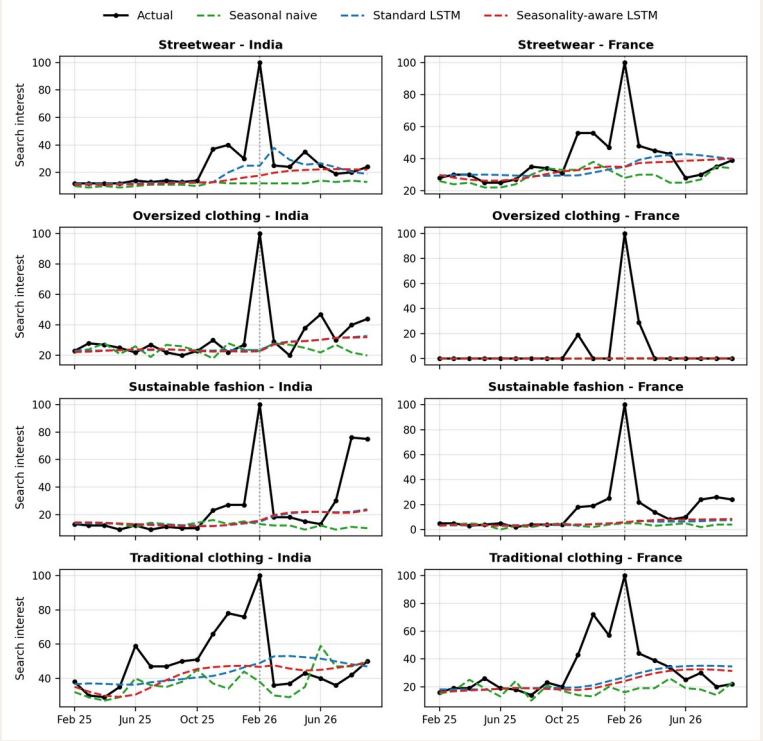
Setup

- Adam, MSE loss, batch 16, up to 100 epochs, early stopping (patience 10); no tuning
- 5 seeds per LSTM (42–46), forecasts averaged; seed variability reported
- Metrics: MAE and RMSE per country and keyword; sensitivity excluding Feb 2026

Results and discussion



- Both LSTMs beat seasonal naive in 7 of 8 series.
- Seasonal features: 10.80 vs 11.00 MAE, within seed noise (~0.5); better in only 3 of 8 series.
- Persistence MAE 11.45: LSTM gain is largely level adaptation. Without Feb 2026 all MAEs fall ~30%; MAE ranking unchanged.



Test 2025-02 to 2026-09 (dotted line: Feb 2026). Series won (MAE): standard 4, seasonal 3, naive 1

Conclusion and future scope

What we found

- LSTMs lower average error than seasonal naive (about 14–16% MAE)
- Sine/cosine month features: no reliable gain over a standard LSTM
- India vs France: descriptive only (8 series, independent normalization)
- Unexplained Feb 2026 peak and Nov 2025 shift limit generality
- Contribution: the comparison and seasonal-feature test, not a new model

Future scope

- Re-collect data; investigate 2025–26 anomalies
- Add persistence, ETS, SARIMA baselines
- Rolling-origin and multi-step evaluation
- Pooled LSTM across series; more keywords, Hindi/French queries
- Statistical testing on larger sets